

MOBILE, AL, USA

WMO: 722230

Lat: 30.688N

Lon: 88.246W

Elev: 215

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	30.9	12.0	10.2	34.4	16.8	12.9	39.2	23.5	59.2	20.6	55.4	6.9	0	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	94.2	77.1	92.3	76.8	90.7	76.4	80.3	88.6	79.4	87.4	78.5	86.1	6.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.9	83.7	77.3	143.2	82.7	76.5	139.2	82.1	43.9	88.6	42.9	87.3	42.1	86.5	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.9	17.7	15.9	DB	21.0	97.1	3.9	2.7	18.1	99.0	15.8	100.6	13.6	102.1	10.7	104.1
			WB	18.6	82.0	3.6	1.8	16.1	83.3	14.0	84.4	12.0	85.4	9.4	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.7	51.0	55.0	60.8	67.1	74.5	80.0	81.8	81.8	78.1	69.0	59.3	53.4
	DBStd	13.05	10.09	9.40	8.28	6.25	4.90	3.17	2.53	2.77	4.24	7.26	8.32	9.67
	HDD50	271	113	51	15	0	0	0	0	0	0	0	16	76
	HDD65	1600	443	297	176	50	5	0	0	0	1	47	207	374
	CDD50	6739	144	191	350	514	760	900	985	985	843	588	296	183
	CDD65	2591	9	17	46	113	299	450	520	520	394	170	37	16
	CDH74	22718	10	43	218	736	2545	4264	5061	5050	3343	1247	171	30
	CDH80	8656	0	0	17	105	857	1773	2143	2150	1291	306	14	0
(14) Wind	WSAvg	7.0	8.2	8.4	8.5	8.1	6.9	5.8	5.2	5.1	6.1	6.6	7.0	7.7
(15) Precipitation	PrecAvg	65.8	5.5	4.5	6.1	5.4	5.0	6.0	8.2	6.5	5.4	3.3	4.5	5.2
	PrecMax	84.7	16.1	10.2	15.7	17.7	14.9	17.8	22.7	12.8	21.7	9.5	11.7	14.5
	PrecMin	42.2	1.0	0.7	0.6	0.6	0.4	1.7	3.9	1.1	2.0	0.0	1.3	1.3
	PrecStd	11.6	3.6	2.4	3.4	4.1	3.2	3.5	3.7	3.0	4.2	2.6	2.9	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	78.3	82.2	85.4	92.5	96.3	96.3	96.5	94.4	89.0	82.0	77.1
		MCWB	66.9	67.2	66.0	69.8	73.3	76.1	78.2	78.0	76.3	73.8	68.9	69.5
	2%	DB	72.5	75.1	79.2	82.9	89.2	93.0	93.4	93.8	91.4	85.9	78.3	74.4
		MCWB	66.0	66.5	66.2	68.8	72.5	75.8	77.7	77.9	75.4	72.0	68.8	68.7
	5%	DB	70.0	72.0	76.4	80.8	87.1	90.7	91.5	91.8	89.3	83.5	75.7	71.9
		MCWB	64.6	65.4	64.9	67.8	72.2	75.7	77.4	77.5	74.9	71.1	67.4	67.3
	10%	DB	67.0	69.4	73.5	78.6	85.0	88.7	89.7	89.9	87.3	81.0	73.1	69.0
		MCWB	62.9	63.8	64.4	66.8	71.5	75.4	77.1	77.0	74.5	69.8	65.7	64.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	71.4	72.3	74.9	77.7	80.4	81.8	82.1	80.2	78.0	74.9	73.5
		MCDB	72.6	74.2	76.0	80.3	85.8	88.7	90.2	91.5	87.5	83.7	77.6	74.8
	2%	WB	68.0	69.4	70.2	73.2	76.3	79.2	80.3	80.3	78.5	76.5	71.9	70.4
		MCDB	70.7	72.1	74.1	78.1	83.4	87.3	88.6	88.8	85.7	81.8	75.0	72.6
	5%	WB	66.0	67.6	68.6	71.8	75.3	78.2	79.4	79.3	77.5	75.0	69.7	68.1
		MCDB	69.0	70.7	72.8	76.4	81.7	85.9	87.4	87.4	84.6	79.9	73.6	70.8
	10%	WB	63.6	65.3	66.9	70.4	74.2	77.3	78.4	78.4	76.7	73.1	67.6	65.7
		MCDB	66.6	68.3	71.3	75.2	80.8	84.7	86.1	86.0	83.3	77.8	71.7	68.7
(35) Mean Daily Temperature Range	MDBR		20.0	19.8	20.5	20.4	19.7	17.3	16.5	16.7	17.7	20.3	21.3	19.2
	5% DB	MCDBR	18.5	18.6	20.3	20.6	21.3	19.8	18.9	19.1	19.6	20.4	20.3	17.8
		MCWBR	12.8	11.3	9.9	9.2	8.0	6.9	6.7	6.5	7.2	9.1	11.7	12.7
	5% WB	MCDBR	15.9	15.1	16.3	15.8	16.8	16.9	16.9	16.8	16.3	15.7	17.2	16.3
		MCWBR	13.6	11.7	10.6	8.9	7.2	6.8	6.9	6.7	6.8	8.6	12.6	13.6
(40) Clear-Sky Solar Irradiance	taub		0.328	0.345	0.369	0.398	0.433	0.460	0.510	0.499	0.443	0.379	0.350	0.336
	taud		2.552	2.507	2.433	2.355	2.287	2.263	2.145	2.170	2.295	2.460	2.483	2.538
	Ebn at Noon		287	292	290	284	273	264	251	252	263	276	276	278
	Edh at Noon		27	31	35	40	43	43	49	47	40	32	29	26
(44) All-Sky Solar Radiation	RadAvg		957	1134	1531	1873	2066	1985	1877	1769	1651	1435	1095	841
	RadStd		75	157	122	150	113	151	126	124	104	150	96	76

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+139
(47) Regional (33 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+164

Nomenclature: See separate page